Rafael Bertolotto Stefanelli

Curriculum Vitae

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†: name translated from Spanish.

Education

Expected 2030 — **PhD degree in Astronomy and Astrophysics, Steward Observatory, University of Arizona**.
Advisor: Kathryne Daniel.

2024 — **Master's degree in Astronomy, Facultad de Ciencias, Universidad de la República**
Thesis title: *Inferring the initial mass function of young co-moving stellar groups in the solar neighborhood*†.
Advisors: Juan José Downes, Genaro Suárez.

2021 — **Bachelor's degree in Astronomy, Facultad de Ciencias, Universidad de la República**
Thesis title: *Detecting young co-moving stellar groups in the solar neighborhood*†.
Advisor: Juan José Downes.

Honors and Awards

2026 — **Recipient of the American Astronomical Society Chambliss Astronomy Achievement Student Award** from the American Astronomical Society for my poster presentation at the American Astronomical Society 247 Winter Meeting.

2023 — **McCarthy-Stoeger fellowship** from the Vatican Observatory.

2022 — **Master's degree Scholarship** from the *Comisión Académica de Posgrado*.

Teaching Experience

2024 — **Galactic and extra-galactic astronomy teaching assistant**, Facultad de Ciencias, Universidad de la República. This is a fourth year undergrad course for astronomy students.

Introduction to computational physics teaching assistant, Facultad de Ciencias, Universidad de la República. This is a third year undergrad course for astronomy and physics students.

2021 - 2023 — **Introduction to the science of Earth and Space II teaching assistant**, Facultad de Ciencias, Universidad de la República. This is a first year undergrad course for astronomy students.

General physics teaching assistant, Facultad de Ciencias, Universidad de la República. This is a first year undergrad course for biology and biochemistry students.

2021 – 2022 — **High school physics teacher,** Lycée Français Jules Supervielles, Montevideo.

Teaching Experience (continued)

2021 **Planetary sciences teaching assistant**, Facultad de Ciencias, Universidad de la República. This is a third year undergrad course for astronomy students.

Skills

Languages	Spanish (native), French (native), Portuguese (native) and English (bilingual)
Coding	Python, R, LaTeX.
Databases	MySQL.
Web Dev	HTML, CSS, JavaScript.
Others	Topcat, Github.

Research Topics

- Galactic dynamics - Galactic structure - Spiral galaxies - Stellar clusters - Clusters in galactic context - Stellar formation - Stellar initial mass function

Scientific Publications

- **Rafael Bertolotto-Stefanelli**, Juan José Downes, Genaro Suárez, Cecilia Mateu, Jonathan Gagné, and Carlos Román Zúñiga; *The stellar initial mass function of nearby young moving groups* (2026); published peer-reviewed article by the Astronomy & Astrophysics journal.

Conferences

2026 **American Astronomical Society 247 Winter Meeting**, where I presented the first results of the first project of my PhD.

2025 **Astrodynamics: From the classroom to the Cosmos**, where I presented my master's thesis main results during a contributed talk.

2024 **Cool Stars 22**, where I presented my master's thesis main results in the form of a poster.

2023 **XVII Latin American Regional IAU Meeting**, where I presented my master's thesis main results during a contribution talk in the division of Star Formation.

XII Friends Of Friends meeting, where I presented a poster about my master's thesis work.

2022 **Annual meeting of the Uruguayan Astronomical Society**†., where I gave a presentation on my ongoing master's degree thesis.

2021 **Annual meeting of the Uruguayan Astronomical Society**†., where I gave a presentation of my bachelor's degree thesis most important results.

2020 **Annual meeting of the Uruguayan Astronomical Society**†., as attendant.

X Workshop on Planetary Sciences†, as attendant.

Conferences (continued)

2017 **Asteroids, Comets and Meteors international conference**, as attendant.

Short courses and internships

2024-2025 One year researching at Steward Observatory (University of Arizona) under the supervision of professor Dennis Zaritsky.

2023 One month at the Vatican Observatory Summer School at Vatican Observatory in Albano Laziale, Italy.

One week internship at the Astronomical Institute of the *Universidad Nacional Autónoma de México* in Ensenada, Baja California, México.

One week internship at the American Museum of Natural History (New York).

2022 One month internship at the Astronomical Institute of the *Universidad Nacional Autónoma de México* in Ensenada, Baja California, México.

Other presentations

2025 **Seminar** at the Astronomy department of the Facultad de Ciencias, Universidad de la República as part of the Astronomy department seminar series.

2024 **Seminar** at NOIRLab / NSF National Optical-IR Astronomy Research Laboratory, as part of the FLASH talk series.

2023 **Colloquium** at the Astronomical Institute of the *Universidad Nacional Autónoma de México* in Ensenada, Baja California, México, as part of the weekly joint colloquium series of the Astronomical Institute of Ensenada†.

2022 **Colloquium** at the Astronomical Institute of the *Universidad Nacional Autónoma de México* in Ensenada, Baja California, México, as part of the weekly joint colloquium series of the Astronomical Institute of Ensenada†.

Positions of responsibility

2023 Member of the Local Organizing Committee for the *XVII Latin American Regional IAU Meeting.*

2022 Member of the Local Organizing Committee for the *Annual meeting of the Uruguayan Astronomical Society*†.

Science Communication and Community Engagement

2025-today Steward Observatory Graduate Student Colloquium Lunch organizer. I organize and moderate weekly lunches with graduate students and the department colloquium speaker, and participate in managing the budget for these events.

2026 Outreach talk at the Splendido retirement community as part of the Splendido Astronomy Lecture Series provided by Steward Observatory, University of Arizona.

2024 Public outreach talk as part of the Astrocharla outreach program at Steward Observatory, University of Arizona.

Science Communication and Community Engagement (continued)

- Outreach talk at the elementary school *Santa Bernardita* for 3rd and 4th grade students on behalf of the Astronomy department, Facultad de Ciencias, Universidad de la República.

References

Kate Daniel	PhD thesis advisor
Juan José Downes	Undergraduate and Master thesis advisor
Dennis Zaritsky	Research advisor